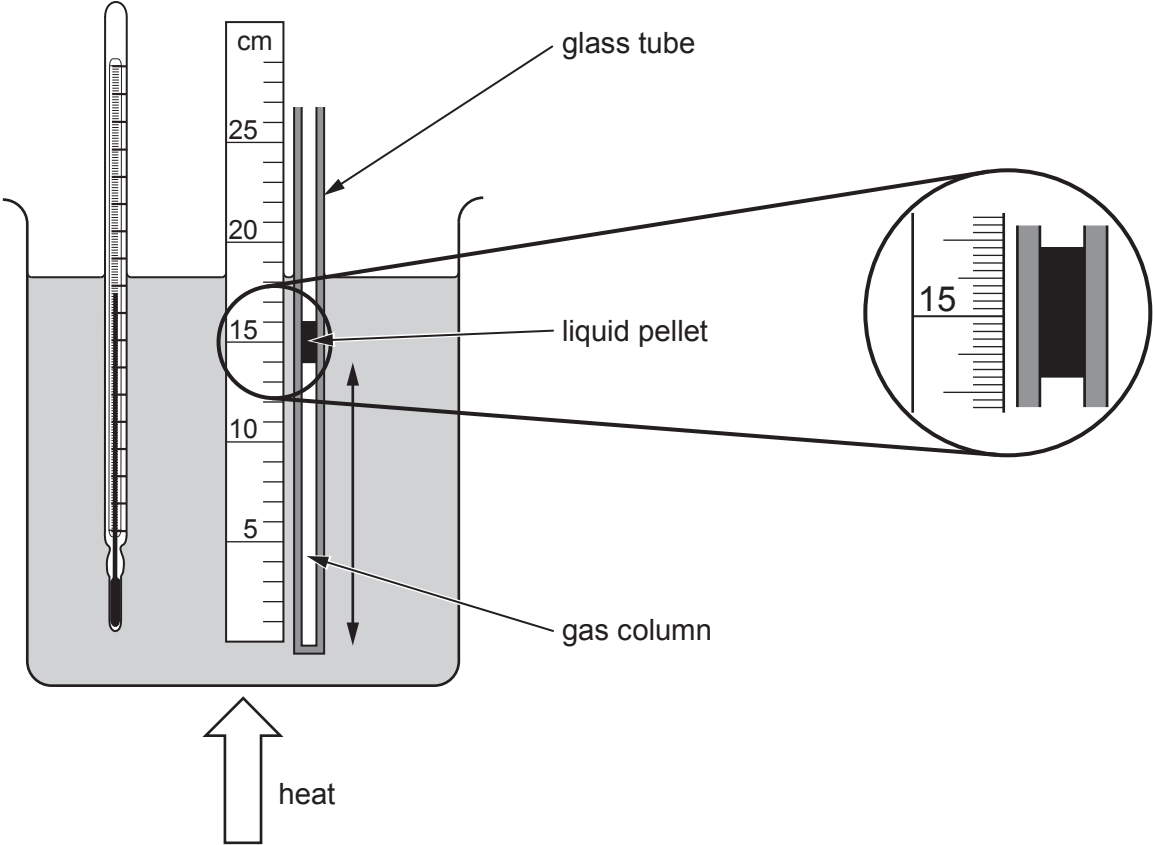


SECTION A

Answer all questions.

1. A student performs an experiment to estimate the Celsius temperature of absolute zero.



The length, l , of the gas column and its temperature, θ , are measured at atmospheric pressure, 1.01×10^5 Pa. The temperature of the water is initially 0.0°C . It is then increased to 80.0°C . Readings are taken at 20.0°C intervals. The scale used to measure the length gives an uncertainty of ± 0.1 cm at the top of the column and an uncertainty of ± 0.1 cm at the bottom.

Values of θ and l are recorded in the table below.

$\theta / ^\circ\text{C}$	l / cm
0.0	11.5
20.0	12.5
40.0	13.2
60.0	14.2
80.0	15.0



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(a) Justify the number of significant figures used to record the length, l . [2]

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(b) The glass tube in which the gas is trapped has an internal diameter of (1.5 ± 0.1) mm.

(i) Calculate the volume of the gas at 0.0°C . [2]

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(ii) Show that the percentage uncertainty in this volume, V , is approximately 15%. [3]

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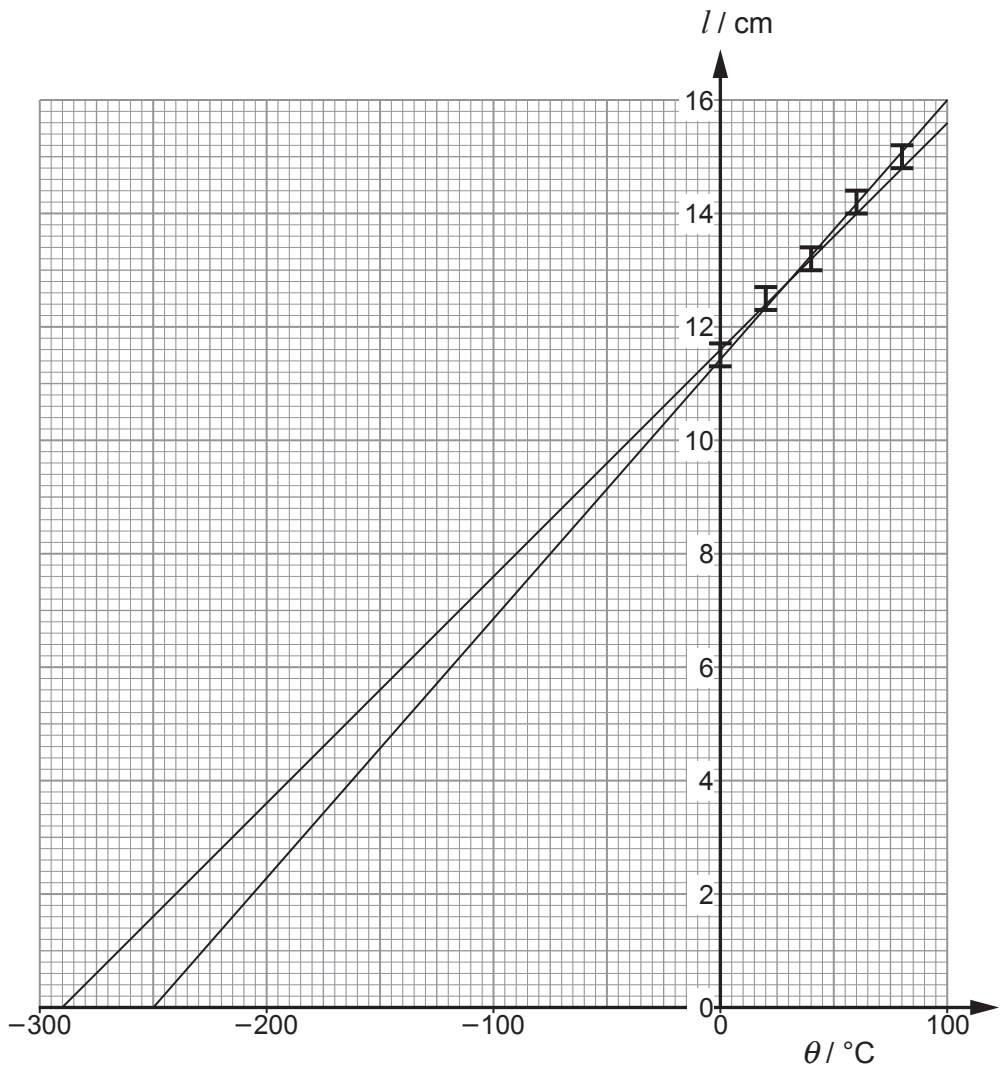
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- (c) The length, l , is plotted against the temperature, θ . Lines of maximum and minimum gradient consistent with the error bars have been drawn.



- (i) Use the graph to estimate the temperature of absolute zero in $^\circ\text{C}$, together with its **absolute** uncertainty. [2]

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(ii) Justify the statement that *the results are consistent with the ideal gas equation*. [4]

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(d) State what happens to the behaviour of the gas molecules as the temperature approaches absolute zero. [1]

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(e) The scale resolution of the ruler affects the final uncertainty in the value of absolute zero. Identify **two** other factors that could affect the accuracy. [2]

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